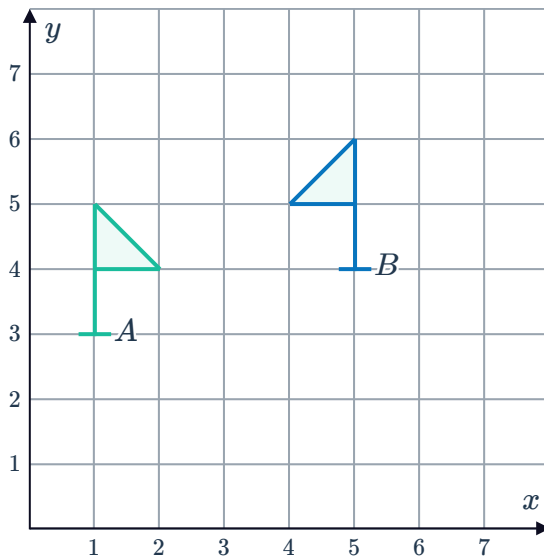


Composing transformations

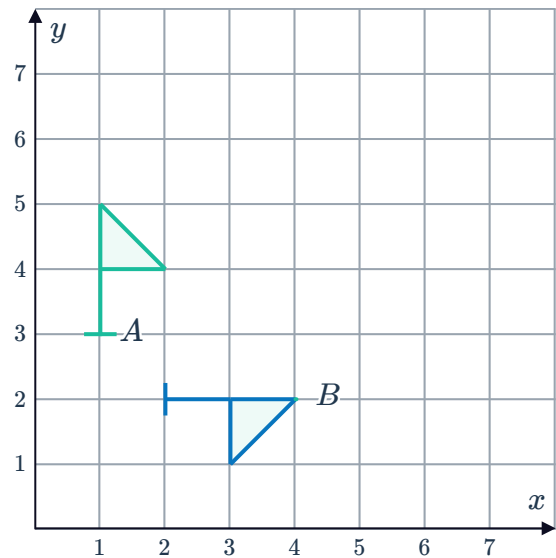


- 1 For each of the following, state the two types of transformations that would be needed to get from Flag A to Flag B:

a

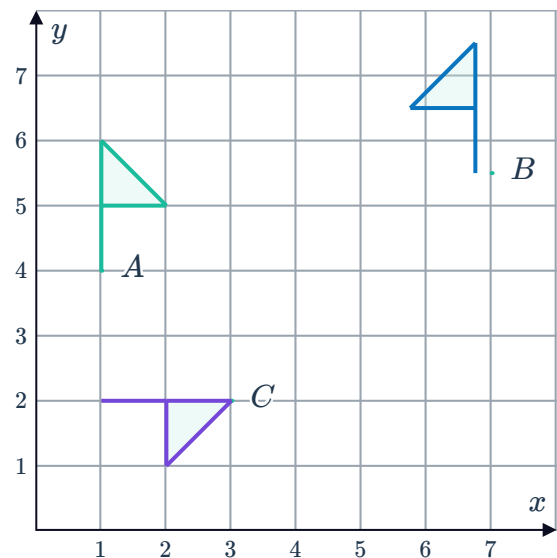


b

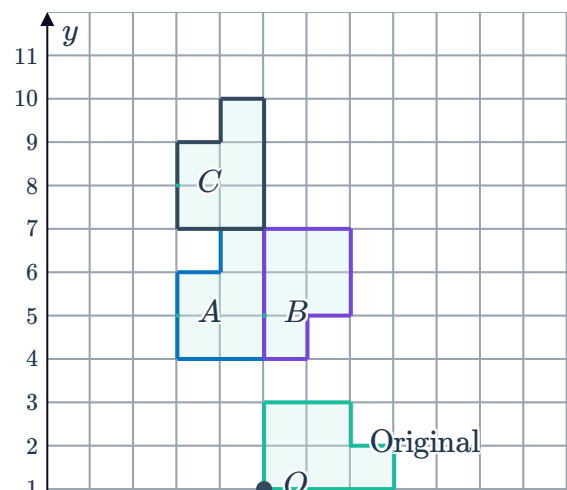


- 2 State the two types of transformations that would be needed to get from:

- a Flag A to Flag B.
b Flag A to Flag C.



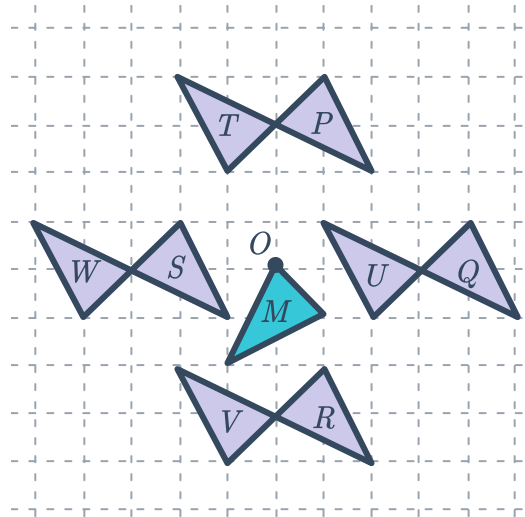
- 3 When the original shape is rotated 90° anti-clockwise about point O and then translated 3 units up, which of the shapes shown is the new shape?



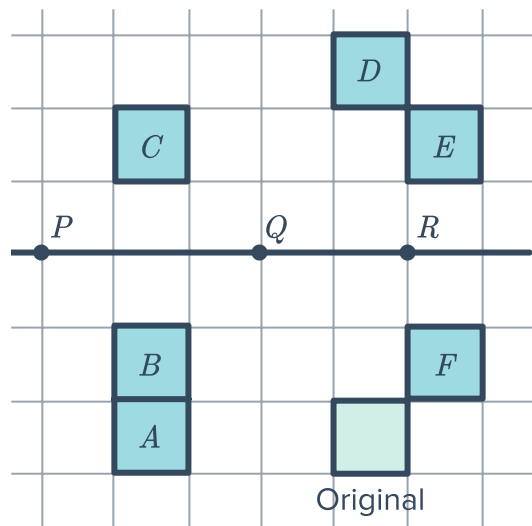


- 4 Triangle M is to be rotated 90° clockwise about point O , then translated 3 units left.

Which triangle corresponds to the result of this transformation?

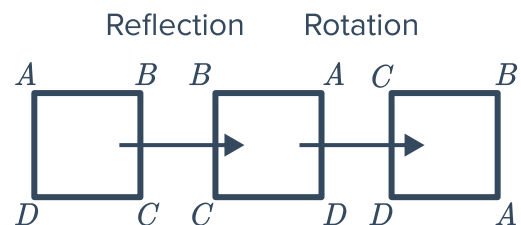


- 5 Identify the correct square when the original is rotated 90° anticlockwise about point Q , then reflected across the line PR , and then translated 4 units to the left.



- 6 Square $ABCD$ is reflected across a vertical line and then rotated 90° clockwise about its centre. This process is repeated with each resulting square. The result of the first set of reflection and rotation is shown:

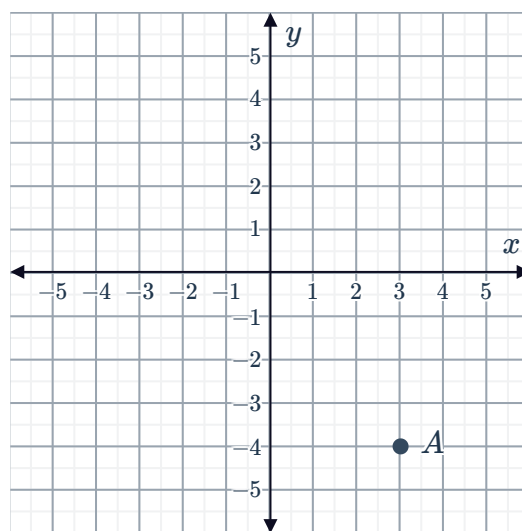
After 13 repetitions of this process, write down the vertices, clockwise from the top left corner.



7 Point A on the plane is to undergo two successive transformations:

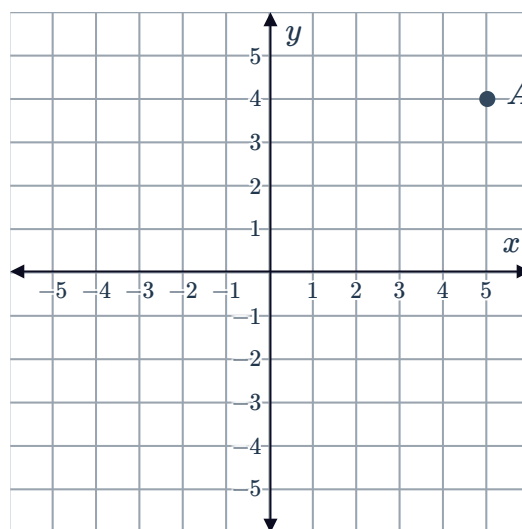


- a Plot point A' , which is the result when point A is translated 4 units left.
- b Plot point A'' , which is the result when point A' is rotated 90° clockwise about the origin.



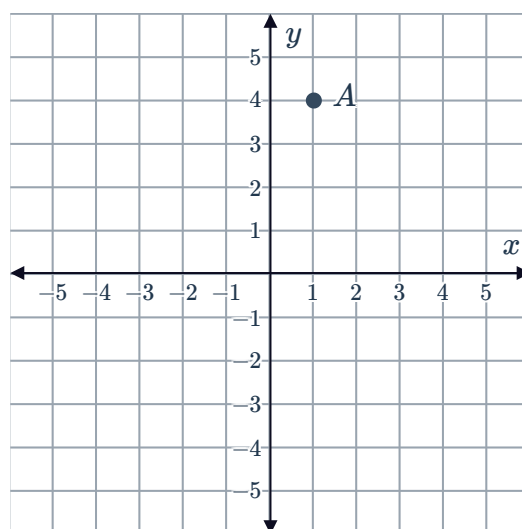
8 Point A on the plane is to undergo two successive transformations:

- a Plot point A' , which is the result when point A is translated 5 units downwards.
- b Plot point A'' , which is the result when point A' is rotated 90° anticlockwise about the origin.

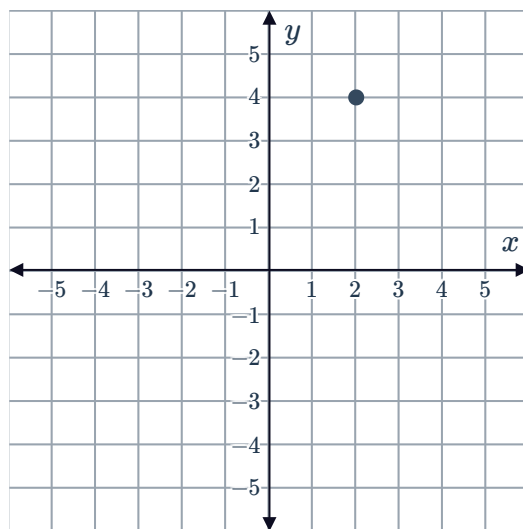


9 Point A on the plane is to undergo two successive transformations:

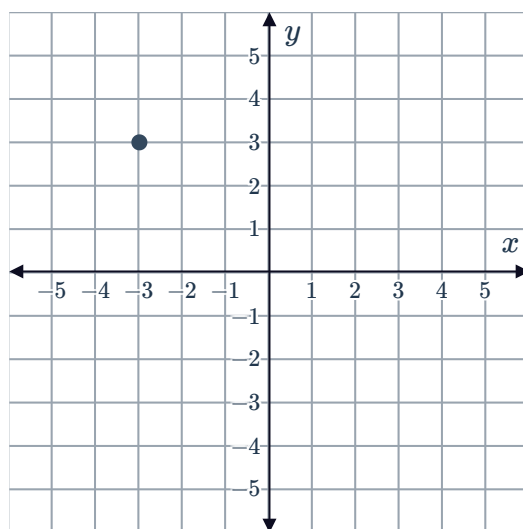
- a Plot point A' , which is the result when point A is translated 3 units right and 5 units down.
- b Plot point A'' , which is the result when point A' is rotated 90° anticlockwise about the origin.



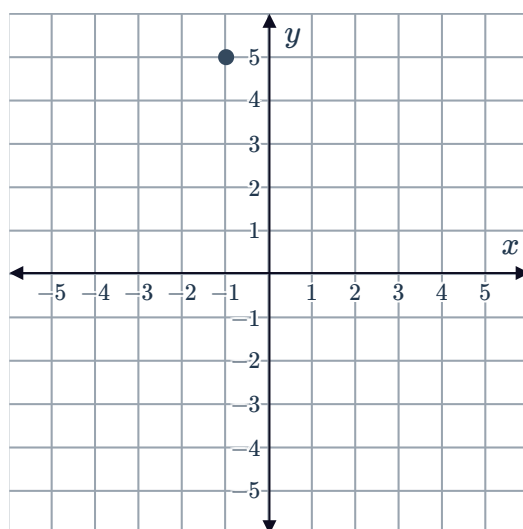
- 10 For the point on the plane, plot the point that will result from a translation of 5 units left followed by a rotation of 90° clockwise about the origin.



- 11 For the point on the plane, plot the point that will result from a translation of 2 units downwards followed by a rotation of 90° anticlockwise about the origin.



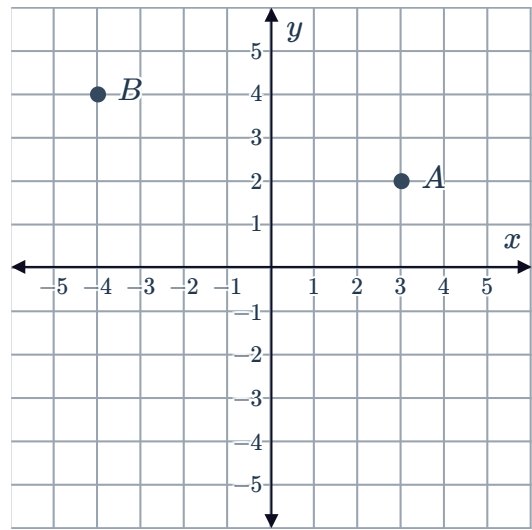
- 12 For the point on the plane, plot the point that will result from a translation of 3 units left and 4 units down, followed by a rotation of 90° anticlockwise about the origin.



- 13 Consider the two points, A and B on the plane:

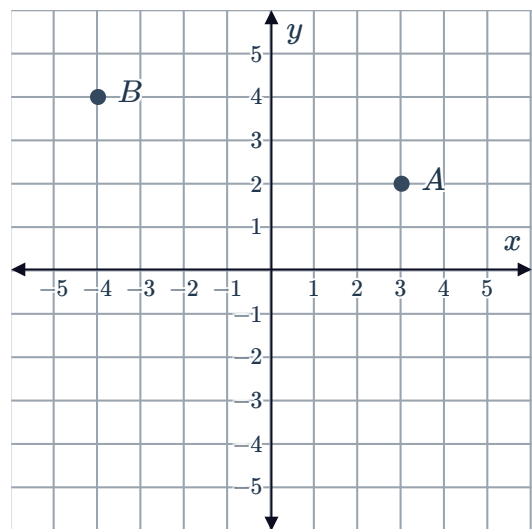


Plot the points that would result when A and B are first rotated 180° about the origin, and then reflected across the x -axis.

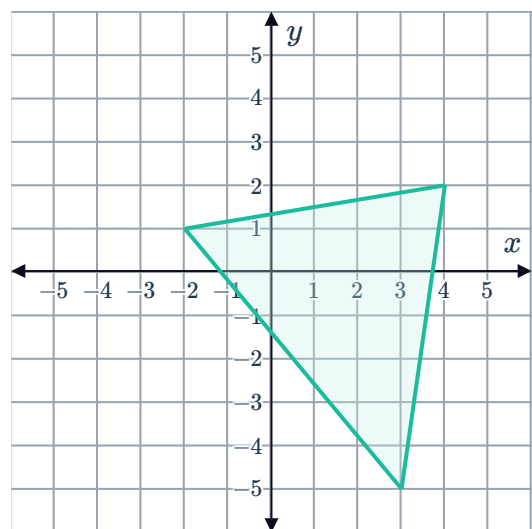


- 14 Consider the two points, A and B on the plane:

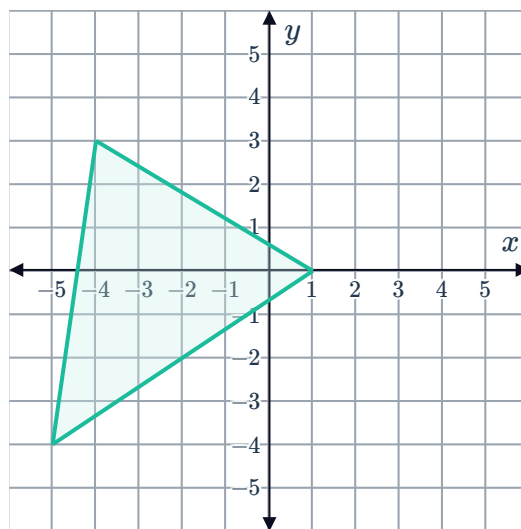
- a Plot the points that would result when A and B are rotated 180° about the origin and reflected across the y -axis.
- b What single transformation would give the same result?



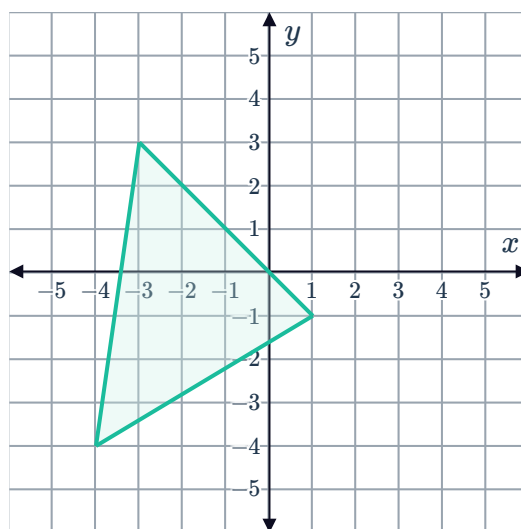
- 15 Plot the triangle that will result when the given triangle is reflected across the x -axis and translated 3 units to the left.



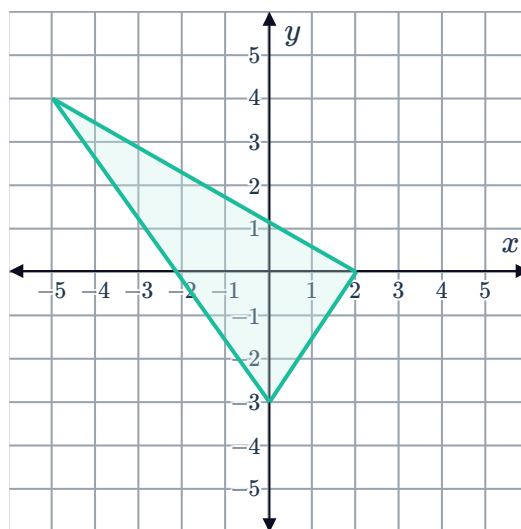
- 16 Plot the triangle that will result when the given triangle is reflected across the y -axis and translated 2 units up.



- 17 Plot the triangle that will result when the given triangle is reflected across the y -axis, and translated 3 units left and 2 units down.



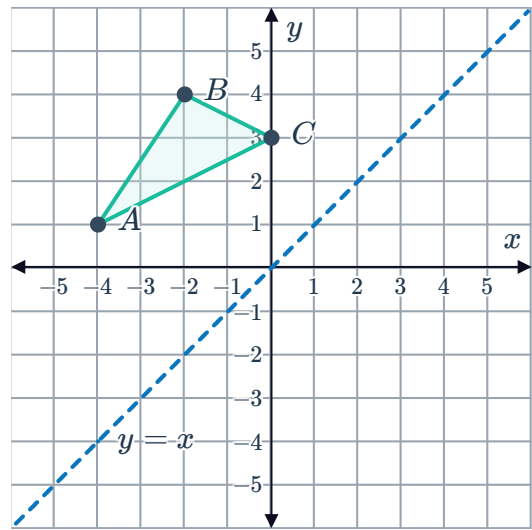
- 18 Plot the triangle that will result when the given triangle is reflected across the x -axis, and translated 3 unit right and 2 units up.



19 Consider the triangle ABC :

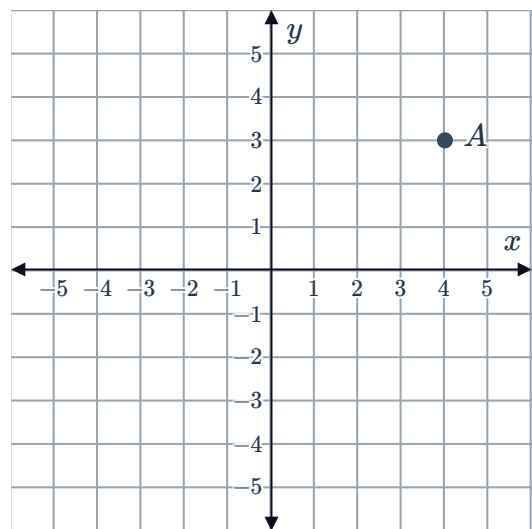


- a Plot triangle $A'B'C'$, the result of reflecting triangle ABC across the line $y = x$.
- b Suppose triangle $A'B'C'$ is reflected across the y -axis to form triangle $A''B''C''$. What single transformation would transform triangle ABC straight to triangle $A''B''C''$?

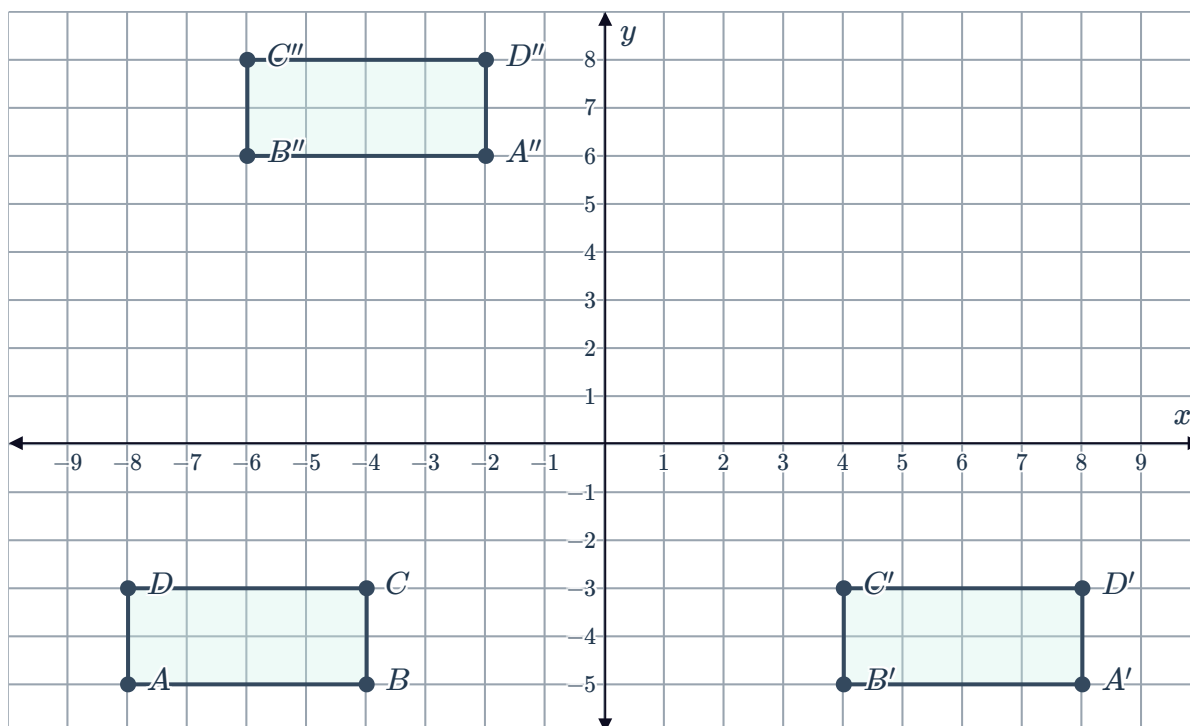


20 Point A is marked on the number plane:

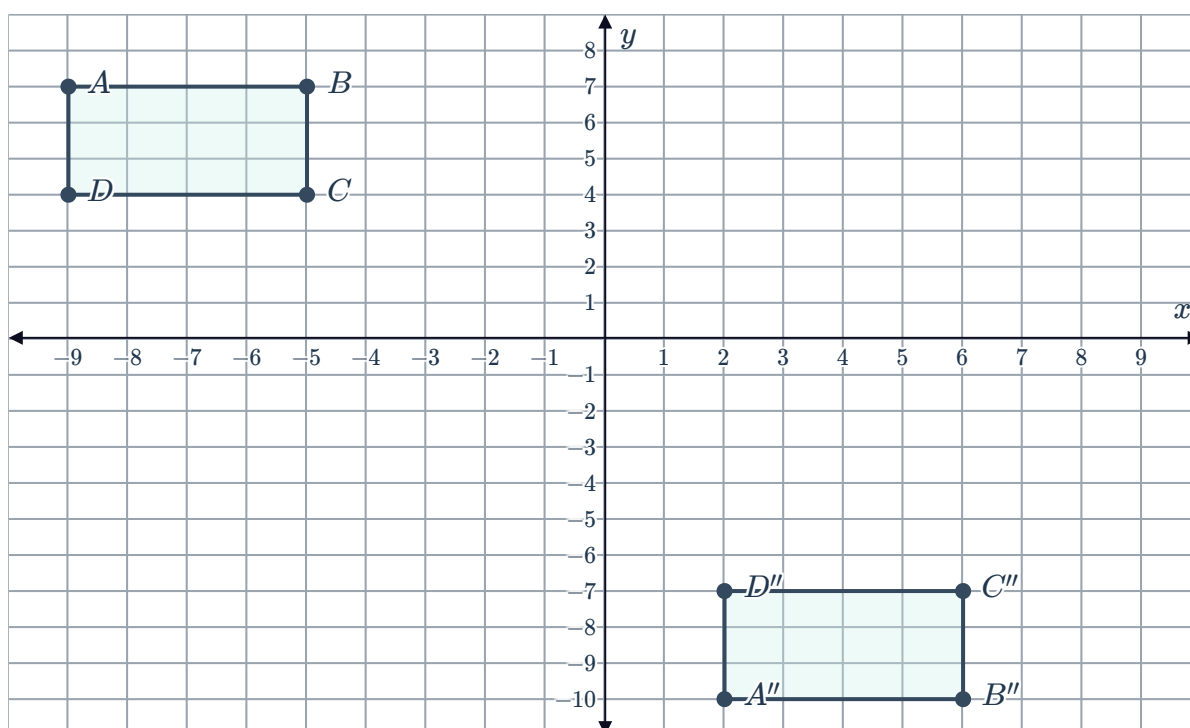
- a If point A is translated 4 units down to make point B , what are the coordinates of point B ?
- b Across what line could point A be reflected across so that it makes point B ?



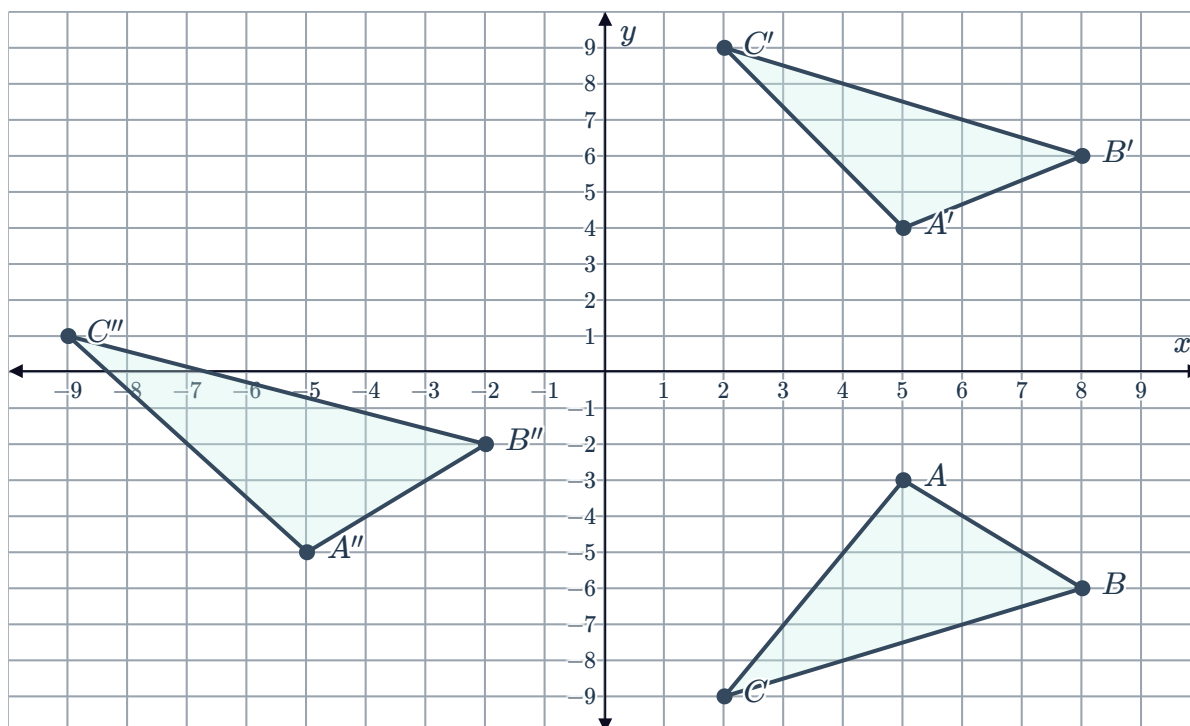
- 21 Describe the sequence of transformations required to get from quadrilateral $ABCD$ to quadrilateral $A''B''C''D''$. Use quadrilateral $A'B'C'D'$ as a guide.



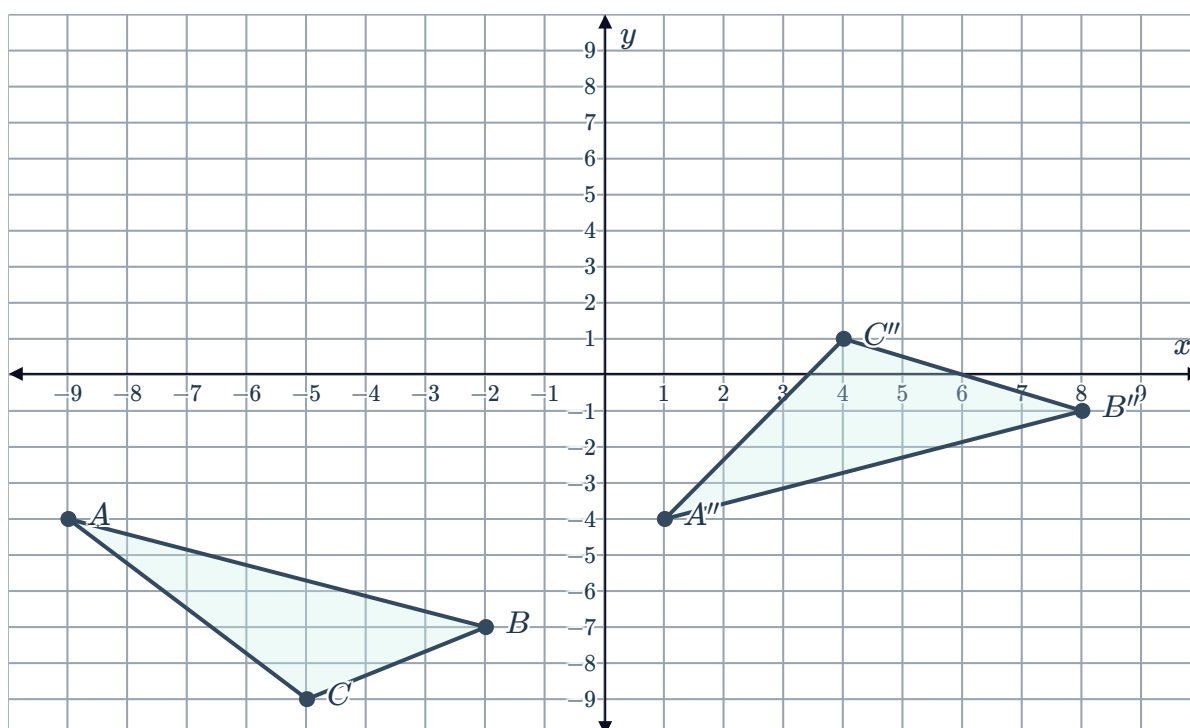
- 22 Describe the sequence of transformations required to get from quadrilateral $ABCD$ to quadrilateral $A''B''C''D''$.



- 23** Describe the sequence of transformations required to get from triangle ABC to triangle $A''B''C''$. Use triangle $A'B'C'$ as a guide.



- 24** Describe the sequence of transformations required to get from triangle ABC to triangle $A''B''C''$.



- 25** Point $A(-10, 6)$ is rotated 90° clockwise about the origin to form point B .
- Write down the the coordinates of point B .
 - Point B is then translated to the left so that it lies on the same vertical line as point A . How many units to the left is point B translated?

translating it 5 units down.



- 36** Points $A(2, -8)$, $B(8, 5)$ and $C(7, 6)$ are the vertices of a triangle. Write down the coordinates A' , B' and C' that result from reflecting the triangle across the y -axis and translating it 4 units right.
- 37** Points $A(-9, -1)$, $B(10, 5)$ and $C(10, 1)$ are the vertices of a triangle. Write down the coordinates A' , B' and C' that result from reflecting the triangle across the y -axis and translating it 4 units down.
- 38** Points $A(6, 7)$, $B(-3, 8)$ and $C(-3, 10)$ are the vertices of a triangle. Write down the coordinates A' , B' and C' that result from reflecting the triangle across the x -axis, and translating it 5 units left and 4 units down.
- 39** Points $A(4, -4)$, $B(10, 3)$ and $C(9, 9)$ are the vertices of a triangle. Write down the coordinates A' , B' and C' that result from reflecting the triangle across the y -axis, and translating it 3 units right and 5 units down.
- 40** For any point (x, y) on the plane, write down the coordinates of the point after a rotation of 180° anticlockwise about the origin.
- 41** What two reflections result in the same transformation as rotating a point 180° anticlockwise about the origin?
- 42** For any point (a, b) on the plane, state the coordinates of the point after a reflection across the x -axis.
- 43** Consider the point $A(-9, 0)$.
- a** If point A is rotated 90° clockwise about the origin to produce point B , find the coordinates of point B .
 - b** Describe the two translations that can be used to move point A to point B .
- 44** Consider the point $A(-5, 6)$.
- a** If point A is reflected across the y -axis to produce point B , and point B is then reflected across the x -axis to produce point C , find the coordinates of point C .
 - b** Describe a single transformation to move point A to point C .